Beyond Numerical Attributes: Semantic-Consistent Metro Flow Forecasting via LLM-Generated Station Descriptions

Anonymous Author(s)

Abstract

Accurate metro flow forecasting is vital for efficient urban transportation management and passenger experience, yet remains challenging due to predetermined scheduling and complex spatiotemporal dependencies over fixed physical infrastructure. Existing efforts typically model metro networks as graphs and learn dependencies from historical flow signals and numerical station attributes while ignoring rich station semantics concerning functionality, location, and urban context, which limits both generalization and interpretability. In this study, we introduce **SCM**, a novel **S**emantic-**C**onsistent **M**etro flow forecasting solution that integrates large language model (LLM)-generated station semantics into spatiotemporal dynamic modeling over the physical metro infrastructure. Specifically, SCM leverages an LLM to generate structured station descriptions encoding geographic context, functional characteristics, and surrounding points of interest, thereby capturing station functions and their urban interactions. Meanwhile, SCM learns station-level interactions via dynamic dependency learning, complemented by multi-scale spatiotemporal convolutions to characterize both short- and long-range flow patterns. We further introduce an information-theoretic semantic consistency objective with a spatial coherence constraint to align station semantics with flow dynamics. Experiments on two real-world metro datasets show that SCM consistently outperforms representative baselines.

1 Introduction

Urban Rail Transit Systems (URTS) [Zhu *et al.*, 2024], particularly metro networks, play a critical role in modern intelligent transportation systems (ITS). Driven by advances in information infrastructures and sensing technologies, large-scale metro passenger flow data have become increasingly available, making accurate metro flow forecasting feasible and essential for effective urban traffic management, improved operational efficiency, and enhanced passenger travel experience. [Xiong *et al.*, 2019; Xie *et al.*, 2023].

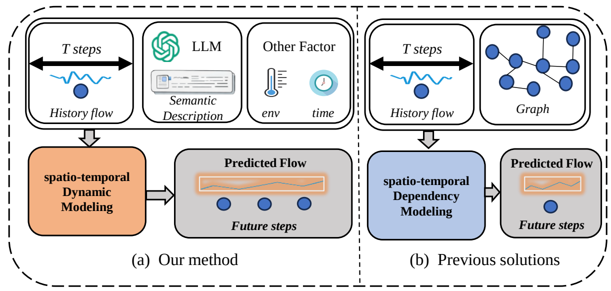

Figure 1: The difference between ours and previous solutions.

In metro systems, metro flow forecasting typically aims to predict the future number of passengers at each station by modeling complex spatiotemporal dynamics, which serves as a vital prerequisite for ensuring supply-demand balance and optimizing resource allocation. Due to intrinsic periodicity, existing traffic flow forecasting paradigms can offer valuable insights for handling metro flow data by capturing higher-order spatiotemporal correlations from historical observations [Liu *et al.*, 2019; Gao *et al.*, 2025]. For instance, (dynamic) graph neural networks (GNNs) are widely used in traffic flow forecasting because they can capture both relational structures and time-varying interactions among sensors. By integrating spatial and temporal signals into unified representations, GNN-based practices achieve improved forecasting accuracy and robustness [Li *et al.*, 2017; Yu *et al.*, 2017]. This capability could be suitable for metro flow data, which shares characteristics with general traffic flow, such as periodic patterns and underlying network topologies.

However, metro flow differs from other urban traffic flows in three fundamental ways due to the unique operational and structural characteristics of metro systems. (1) Stations have fixed locations determined by surrounding functional areas, such as residential, commercial, or transportation hubs, with minimal changes over time, resulting in distinct, station-specific periodic flow patterns. (2) Although the network topology is fixed, inter-station interactions vary across weekdays, weekends, and peak hours, producing time-varying flow relationships. (3) Metro operations follow highly standardized regimes, with train departures and dispatch strictly adhering to predefined timetables, leading to higher predictability and stability compared to other traffic flows.

These distinctive properties of metro flow highlight three key aspects for modeling inter-station interactions and flow dynamics: (C1) *Semantic Station Roles*. Each metro station generally maintains a stable functional role within the city, which corresponds to distinct flow characteristics. For example, stations in business districts, residential areas, or transportation hubs exhibit different flow patterns and periodicities. Traditional methods, including metro flow-specific ones [Xie *et al.*, 2023; Gao *et al.*, 2025], however, struggle to extract and leverage these semantic features, which are crucial for understanding fundamental metro flow dynamics beyond numerical attributes. (C2) *Temporal Inter-Station Correlations.* Although metro stations occupy fixed physical locations, inter-station passenger flow exhibits pronounced spatiotemporal dynamics. For example, strong correlations arise between residential and business district stations during weekday commuting peaks, whereas on weekends, these dynamic correlations often shift toward residential and commercial or entertainment district stations. External factors such as weather and holidays further modulate these patterns, requiring models that adaptively capture station-to-station relationships across spatiotemporal scales. (C3) *Flow-Driven Interaction Dynamics.* Changes in passenger volumes can themselves drive the evolution of inter-station interactions. For example, a sudden surge at one station often triggers adjustments in adjacent stations, creating a chain reaction. This process exhibits complex, time-varying characteristics, requiring models to capture both instantaneous flow changes and dynamically adjust inter-station association strengths.

To address the above concerns, we propose **SCM**, a novel **S**emantic-**C**onsistent **M**etro flow forecasting framework that incorporates large language model (LLM)-generated station semantics into spatiotemporal dynamic modeling over the physical metro infrastructure. Unlike prior methods that rely on static but numerical spatial structures (cf. Fig. 1), SCM first leverages an LLM to generate structured station semantic descriptions, encoding geographic context, functional characteristics, and surrounding points of interest (POIs), which serve as a robust foundation for downstream spatiotemporal learning over numerical metro flow signals (addressing C1). To uncover dynamic relationships between stations (C2), we devise a contextual spatial embedding block, primarily seeking to adaptively fuse interactions among flow, external factors, and semantic descriptions. To capture higher-order spatiotemporal interactions (C3), we then introduce a dynamic signal exploration module that learns station-level interactions via dynamic dependency modeling, complemented by a multi-scale convolution to distill both short- and long-range flow patterns. In addition, an information-theoretic semantic consistency objective is involved with a spatial coherence constraint to jointly refine station semantics and flow dynamics, reducing semantic–spatiotemporal misalignment. In summary, we make the following contributions:

- We propose SCM, a novel semantic-consistent forecasting framework that, for the first time, exploits LLMs to bridge high-level station-centric semantics with low-level numerical flow signals, facilitating unified semantic–spatiotemporal modeling over metro infrastructures.
- Beyond unifying numerical attributes with semantic descriptions, SCM captures station-level interactions via dynamic dependency learning and multi-scale spatiotemporal convolutions for both short- and long-range flow patterns. An information-theoretic semantic consistency objective with a spatial coherence constraint is further introduced to align station semantics with flow dynamics.
- Experiments on two real-world metro datasets show that SCM outperforms representative baselines, uncovering its robustness and effectiveness across diverse urban contexts.

2 Preliminaries

Definition 1 (Metro Network)**.** *Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph representing the metro network, with $|\mathcal{V}| = n$ nodes (e.g., metro stations/sensors) and edge set $\mathcal{E}$ denoting the transit lines. The connectivity structure is formalized by the adjacency matrix $\mathcal{A} \in \mathbb{R}^{n\times n}$, defined such that $\mathcal{A}_{ij} = 1$ if an edge exists between v_i and v_j, and 0 otherwise.*

Definition 2 (Metro Flow)**.** *Let $x_{i,t} \in \mathbb{R}^c$ denote the current flow of node v_i at time step t. The metro flow data for node v_i over observation window ω is defined as $X_i = [x_{i,1}, \ldots, x_{i,\omega}]^\top \in \mathbb{R}^{\omega\times c}$. Aggregating these matrices across all n nodes yields the global flow observations $\mathcal{X} \in \mathbb{R}^{n\times\omega\times c}$.*

Definition 3 (Task: Metro Flow Forecasting)**.** *Given the historical metro flow records $\boldsymbol{\mathcal{X}}_{t-\omega+1:t} \in \mathbb{R}^{n\times c\times\omega}$ and station-specific semantic attributes $\boldsymbol{\mathcal{S}}$, we aim to learn a forecasting model $\boldsymbol{\mathcal{M}}$ with parameters Θ to forecast the metro flow for the subsequent ω time steps. We incorporate external contexts, including date information $\boldsymbol{\mathcal{D}}$ and weather conditions $\boldsymbol{\mathcal{W}}$, to enhance prediction accuracy. The forecasting problem is formulated as learning a mapping function:*

$$\hat{\boldsymbol{Y}}_{t+1:t+\omega} = \boldsymbol{\mathcal{M}}_\Theta(\boldsymbol{\mathcal{X}}_{t-\omega+1:t}|\boldsymbol{\mathcal{S}}, \boldsymbol{\mathcal{D}}, \boldsymbol{\mathcal{W}}), \tag{1}$$

where $\hat{\boldsymbol{Y}}_{t+1:t+\omega} \in \mathbb{R}^{n\times c\times\omega}$ denotes the predicted flow.

3 Methodology

Fig. 2 illustrates the overall architecture of SCM, comprising five core components: Semantic Description Generation, Contextual Spatial Embedding, Dynamic Graph Convolution, Multi-Scale Temporal Convolution, and Task Adaptation, which are described in detail below.

3.1 Semantic Description Generator (SDG)

In our solution, the devised SDG strives to utilize raw station-specific semantic attributes $\boldsymbol{\mathcal{S}}$ to characterize each station. Unlike conventional learnable random embeddings, these features naturally capture inter-station dependencies and flow patterns. In detail, we first extract the refined attributes $\boldsymbol{\mathcal{S}}_i'$ from the raw descriptions using GPT-5-nano, described as:

$$\boldsymbol{\mathcal{S}}_i' = \mathrm{LLM}(\boldsymbol{\mathcal{S}}_i; \boldsymbol{\mathcal{T}}). \tag{2}$$

Herein, $\boldsymbol{\mathcal{T}}$ represents the *prompt* and $\boldsymbol{\mathcal{S}}_i$ comprises multi-dimensional textual descriptions of the i-th station (detailed in Appendix B). The generated semantic outputs $\boldsymbol{\mathcal{S}}_i'$ contain four sections: *(S1) Overview*: Describing urban spatial position and typical metro flow characteristics. *(S2) Functional Features*: Explaining how surrounding POI types influence metro flow. *(S3) Flow Explanation*: Analyzing geographic

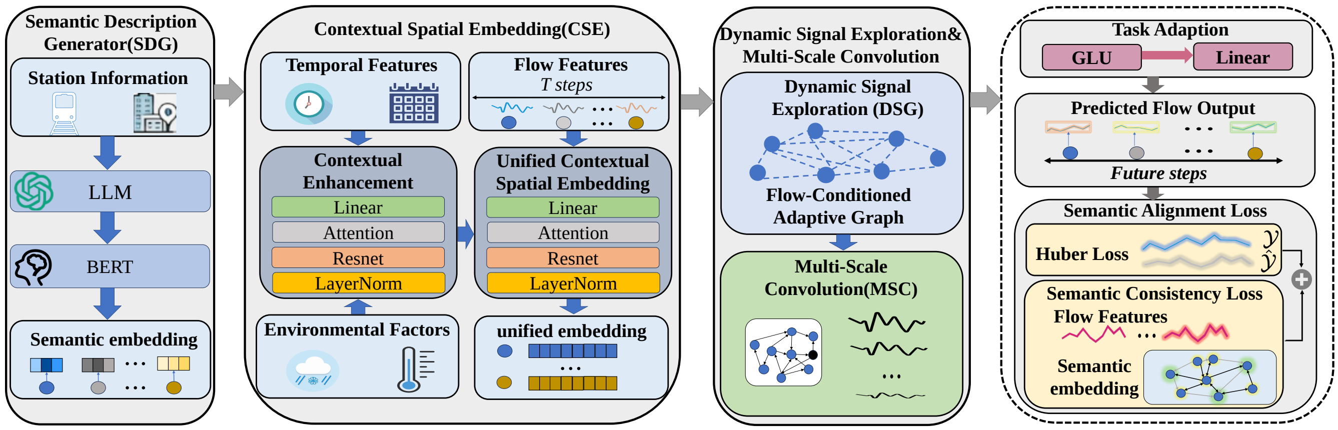

Figure 2: The overall architecture of our proposed SCM.

and functional characteristics for peak period inference. *(S4) Semantic Tags*: Providing key descriptors (e.g., transfer Hub, commercial area, and residential zone).

To achieve dimensional alignment, we employ a pretrained BERT [Devlin *et al.*, 2018] to transform $\boldsymbol{\mathcal{S}}'_i$ into embeddings:

$$\boldsymbol{e}_i = \mathrm{BERT}(\boldsymbol{\mathcal{S}}'_i) = \frac{1}{|\boldsymbol{\mathcal{S}}'_i|} \sum_{j=1}^{|\boldsymbol{\mathcal{S}}'_i|} \boldsymbol{h}_{i,j} \in \mathbb{R}^{d_e}, \tag{3}$$

where $\boldsymbol{h}_{i,j}$ represents the hidden state of the j-th token in the refined attributes $\boldsymbol{\mathcal{S}}'_i$ of station i, and $\boldsymbol{e}_i$ is the resulting semantic embedding. For the entire metro network with n stations, we concatenate these outputs into unified representation $\boldsymbol{\mathcal{S}}^e$:

$$\boldsymbol{\mathcal{S}}^e = [\boldsymbol{e}_1; \boldsymbol{e}_2; \ldots; \boldsymbol{e}_n]^\top \in \mathbb{R}^{n \times d_e}. \tag{4}$$

3.2 Contextual Spatial Embedding (CSE)

CSE aims to integrate station-level semantics with temporal and environmental contexts and aligns the fused representations with flow dynamics. It employs a progressive cross-attention fusion strategy, allowing semantics to selectively attend to contextual signals before interacting with flow.

Contextual Enhancement. As expected, metro flow shows strong temporal regularities, e.g., daily commuting cycles and weekly periodicity. To explicitly model such temporal context, it is necessary to enhance the station-aware representation with date context (e.g., hour-of-day, day-of-week, workday indicators). We thus project the temporal features into the same latent space, which can be described as follows:

$$\boldsymbol{\mathcal{D}}^e = \mathrm{broad}(\mathrm{Linear}(\mathcal{R}(\boldsymbol{\mathcal{D}}))) \in \mathbb{R}^{n \times d_e}, \tag{5}$$

where broad() is the *Broadcast* operation and $\mathcal{R}$ is the *Reshape* operation. We then use a cross-attention mechanism to allow station semantics to adaptively attend to flow patterns:

$$\boldsymbol{Q}^{d,h} = \boldsymbol{\mathcal{S}}^e \boldsymbol{W}_Q^{d,h}, \boldsymbol{K}^{d,h} = \boldsymbol{\mathcal{D}}^e \boldsymbol{W}_K^{d,h}, \boldsymbol{V}^{d,h} = \boldsymbol{\mathcal{D}}^e \boldsymbol{W}_V^{d,h}, \tag{6}$$

where $\boldsymbol{W}_*^{d,h} \in \mathbb{R}^{d_e \times d_e}$ are learnable matrices. For a head $h \in [1, 2, \cdots, H]$, the cross-attention operation is defined as:

$$\boldsymbol{O}^{d,h} = \mathrm{Softmax}(\frac{\boldsymbol{Q}^{d,h}(\boldsymbol{K}^{d,h})^\top}{\sqrt{d_e}})\boldsymbol{V}^{d,h}. \tag{7}$$

The outputs of all heads are then concatenated and projected to obtain the unified representation:

$$\boldsymbol{O}^d = [\boldsymbol{O}^{d,1}; \boldsymbol{O}^{d,2}; \ldots; \boldsymbol{O}^{d,h}]. \tag{8}$$

For notational convenience, we define the aforementioned operation on $\boldsymbol{\mathcal{D}}^e$ and $\boldsymbol{\mathcal{S}}^e$ as MHA$(\cdot)$, i.e., $O^d = \mathrm{MHA}(\boldsymbol{\mathcal{D}}^e, \boldsymbol{\mathcal{S}}^e)$. To stabilize training and preserve the original semantic information, we further adopt a residual connection followed by layer normalization, yielding the final representation:

$$\boldsymbol{\mathcal{O}}^d = \mathrm{LayerNorm}(\boldsymbol{O}^d + \boldsymbol{\mathcal{S}}^e) \in \mathbb{R}^{n \times d_e}, \tag{9}$$

where LayerNorm$(\cdot)$ is the layer normalization operation.

Given that environmental factors significantly impact passenger behavior, we incorporate them as contextual information. We first perform a dimension alignment by reshaping and projecting the weather condition tensor $\boldsymbol{\mathcal{W}} \in \mathbb{R}^{\omega \times c_w}$ (c_w refers to different weather condition) into the latent space:

$$\boldsymbol{\mathcal{W}}^e = \mathrm{broad}(\mathrm{Linear}(\mathcal{R}(\boldsymbol{\mathcal{W}}))) \in \mathbb{R}^{n \times d_e}. \tag{10}$$

The resulting environmental latent tensor is then fused with station-specific attributes via MHA$(\cdot)$, it can be denoted as:

$$\boldsymbol{O}^w = \mathrm{MHA}(\boldsymbol{\mathcal{W}}^e, \boldsymbol{\mathcal{S}}^e), \tag{11}$$

$$\boldsymbol{\mathcal{O}}^w = \mathrm{LayerNorm}(\boldsymbol{\mathcal{S}}^e + \boldsymbol{O}^w) \in \mathbb{R}^{n \times d_e}. \tag{12}$$

Finally, we fuse the embeddings extracted by the above two context-aware blocks and obtain the final output $\boldsymbol{\mathcal{O}}$:

$$\boldsymbol{\mathcal{O}} = \mathrm{LayerNorm}(\boldsymbol{\mathcal{O}}^d + \boldsymbol{\mathcal{O}}^w) \in \mathbb{R}^{n \times d_e}. \tag{13}$$

Unified Contextual Spatial Embedding. Through the progressive fusion of semantic, date, and environmental context, the final contextual spatial embedding $\boldsymbol{\mathcal{O}}$ captures both stable station semantics and dynamically evolving contextual factors. To effectively combine flow observations with contexts, we first reshape the origin flow tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{n \times \omega \times c}$ into a unified dimension, which can be denoted as follows:

$$\boldsymbol{\mathcal{X}}' = \mathrm{Linear}(\mathcal{R}(\boldsymbol{\mathcal{X}})) \in \mathbb{R}^{n \times d_e}. \tag{14}$$

We then use cross-attention between the context signal $\boldsymbol{\mathcal{O}}$ and flow tensors $\boldsymbol{\mathcal{X}}'$, summarized as follows:

$$\boldsymbol{\mathcal{U}} = \mathrm{LayerNorm}(\mathrm{MHA}(\boldsymbol{\mathcal{X}}', \boldsymbol{\mathcal{O}}) + \boldsymbol{\mathcal{O}}), \tag{15}$$

where $\boldsymbol{\mathcal{U}} \in \mathbb{R}^{n \times d_e}$ integrates contextual signals with station-wise flow patterns, serving as the basis for generating the Laplacian matrix, as detailed in the subsequent section.

3.3 Dynamic Signal Exploration (DSG)

To capture dynamic interactions between stations under the evolving metro flow, our DSG is designed to explicitly model time-varying spatial dependencies based on contextual station representations. As revealed by previous efforts [Bai *et al.*, 2020; Lan *et al.*, 2022], the spatial interaction in the latent space between two stations can be effectively characterized by the similarity of their contextual embeddings. Accordingly, we compute a data-driven adjacency matrix by measuring pairwise interactions between station representations:

$$\tilde{\boldsymbol{\mathcal{A}}} = \text{Softmax}(\text{ReLU}(\boldsymbol{\mathcal{U}} \cdot \boldsymbol{\mathcal{U}}^\top)), \tag{16}$$

where $\tilde{\boldsymbol{\mathcal{A}}} \in \mathbb{R}^{b\times n\times n}$ and b refers to the collection size. Based on the learned adjacency matrix, like most prior efforts using graph convolution, we use a Chebyshev polynomial-based graph convolution [Defferrard *et al.*, 2016] to capture higher-order spatial dependencies. The k-th order graph support is recursively defined as:

$$\boldsymbol{T}_0 = I,\ \boldsymbol{T}_1 = \tilde{\boldsymbol{\mathcal{A}}},\ \boldsymbol{T}_k = 2\tilde{\boldsymbol{\mathcal{A}}}\boldsymbol{T}_{k-1} - \boldsymbol{T}_{k-2},\ k \geq 2. \tag{17}$$

Herein, I is the identity matrix and k is the order of the Chebyshev polynomials. Typically, conventional Chebyshev-based graph convolutions use shared convolution parameters across all flow observations, which may limit the ability to capture diverse aggregation patterns under dynamic traffic conditions. To alleviate this limitation, we introduce a node-adaptive convolution parameter selection strategy, where convolution weights are dynamically generated based on contextual station embeddings. Given the contextual station representation $\boldsymbol{\mathcal{U}}$, the convolution weights are selected from a learnable parameter pool via:

$$\boldsymbol{W} = \boldsymbol{\mathcal{U}} \cdot \boldsymbol{W}_p \in \mathbb{R}^{b\times n\times k\times d_i\times d_o}, \tag{18}$$

$$\boldsymbol{b} = \boldsymbol{\mathcal{U}} \cdot \boldsymbol{b}_p \in \mathbb{R}^{b\times n\times d_o}, \tag{19}$$

where $\boldsymbol{W}_p \in \mathbb{R}^{d_e\times k\times d_i\times d_o}$ is the parameter pool of graph convolution kernels, $\boldsymbol{b}_p \in \mathbb{R}^{d_e\times d_o}$ is a learnable bias pool. d_i and d_o are the input and output feature dimensions.

Next, given the flow features $\boldsymbol{\mathcal{X}}'$, the dynamic graph convolution is finally formulated as follows:

$$\boldsymbol{\mathcal{C}} = \pi(\sum_{i=0}^{k-1} \boldsymbol{T}_i \boldsymbol{\mathcal{X}}' \boldsymbol{W}_i + \boldsymbol{b}) \in \mathbb{R}^{b\times n\times d_e}, \tag{20}$$

where $\pi(\cdot)$ is the activation function, implemented as *GELU*.

3.4 Multi-Scale Convolution (MSC)

Our MSC module is designed to integrate spatial information and generate station-specific representations that effectively encapsulate the contextual spatial dependencies. However, metro flow forecasting also needs to model how such spatial representations evolve over future time steps. To this end, we introduce a lightweight multi-scale temporal convolution module to capture temporal patterns at different ranges.

We first replicate the spatial representation along the forecasting horizon to construct a temporal sequence. We then reshape the tensor to apply temporal convolutions independently for each station, which can be summarized as:

$$\boldsymbol{z} = \text{Permute}(\mathcal{R}(\text{Repeat}(\boldsymbol{\mathcal{C}}))) \in \mathbb{R}^{n\times\omega\times d_e}, \tag{21}$$

where each row corresponds to the temporal sequence of a single station. To capture temporal dependencies with different receptive fields, we apply multiple parallel 1D convolutions along the time dimension, denoted as:

$$\boldsymbol{z}_i' = \text{Conv1d}_i(z). \tag{22}$$

Herein, the $\boldsymbol{z}_i'$ is the intermediate temporal feature maps extracted by the 1D convolution with kernel size i. The resulting features are concatenated and projected to hidden dimension:

$$\boldsymbol{\mathcal{Z}} = \mathcal{R}(\text{Linear}(\text{Concat}[\boldsymbol{z}_1'; \boldsymbol{z}_2'; \cdots; \boldsymbol{z}_n'])) \in \mathbb{R}^{n\times\omega\times d_e}. \tag{23}$$

3.5 Task Adaption

Inspired by gating strategies commonly used in flow forecasting and temporal modeling [Yao *et al.*, 2019], we introduce a lightweight Gated Transformation Unit (GTU) to regulate nonlinear information propagation. The GTU computes a gating signal and a nonlinear update as:

$$\boldsymbol{p} = \sigma(\text{Linear}(\boldsymbol{\mathcal{Z}})), \boldsymbol{q} = \tanh(\text{Linear}(\boldsymbol{\mathcal{Z}})). \tag{24}$$

The gate $\boldsymbol{p}$ controls the degree to which new nonlinear information is introduced, while $\boldsymbol{q}$ models candidate updates. An element-wise interpolation between the original representation and the nonlinear update yields the gated output:

$$\boldsymbol{\mathcal{H}} = \boldsymbol{p} \odot \boldsymbol{q} + (1 - \boldsymbol{p}) \odot \boldsymbol{\mathcal{Z}}. \tag{25}$$

Finally, a linear projection is applied to map the gated representations to the target flow dimension, denoted as follows:

$$\hat{\boldsymbol{Y}} = \mathcal{R}(\text{Linear}(\boldsymbol{\mathcal{H}})) \in \mathbb{R}^{n\times w\times c}. \tag{26}$$

3.6 Semantic Alignment Loss

In this study, our goal is to learn a context representation $\boldsymbol{\mathcal{O}}$ that maximizes the mutual information (MI) with the historical flow representation $\boldsymbol{\mathcal{X}}'$ (simplified as $\boldsymbol{\mathcal{X}}$). This ensures that the learned flow features are consistent with the environmental context. Thus, the objective is formulated as:

$$\max_{\theta} \mathcal{I}(\boldsymbol{\mathcal{O}}; \boldsymbol{\mathcal{X}}), \tag{27}$$

where $\mathcal{I}$ denotes the mutual information. By definition, MI quantifies the reduction in uncertainty of $\boldsymbol{\mathcal{X}}$ given $\boldsymbol{\mathcal{O}}$:

$$\begin{aligned} \mathcal{I}(\boldsymbol{\mathcal{O}}; \boldsymbol{\mathcal{X}}) &= \sum_{\boldsymbol{o},\boldsymbol{x}} p(\boldsymbol{o}, \boldsymbol{x}) \log \frac{p(\boldsymbol{o}, \boldsymbol{x})}{p(\boldsymbol{o})p(\boldsymbol{x})} \\ &= \mathbb{E}_{(\boldsymbol{o},\boldsymbol{x})\sim p(\boldsymbol{o},\boldsymbol{x})}[\log \frac{p(\boldsymbol{x}|\boldsymbol{o})}{p(\boldsymbol{x})}]. \end{aligned} \tag{28}$$

However, directly maximizing this quantity is intractable as the true posterior distribution $p(\boldsymbol{x}|\boldsymbol{o})$ is unknown. To address this, we derive a lower bound using InfoNCE. We use a scoring function $f(\boldsymbol{o}, \boldsymbol{x})$ to model the density ratio between the joint distribution and the product of marginals. The optimal value of this function approximates the log-density ratio:

$$f(\boldsymbol{o}, \boldsymbol{x}) \propto \log \frac{p(\boldsymbol{x}|\boldsymbol{o})}{p(\boldsymbol{x})}. \tag{29}$$

In our implementation, we define this scoring function as the temperature-scaled cosine similarity:

$$f(\boldsymbol{o}, \boldsymbol{x}) = \frac{\mathrm{sim}(\boldsymbol{o}, \boldsymbol{x})}{\tau}, \tag{30}$$

where τ is a temperature hyperparameter. By treating the problem as a contrastive learning task – identifying the positive pair $(\boldsymbol{o}_i, \boldsymbol{x}_i)$ from a batch containing $n-1$ other stations, i.e., negative pairs $(\boldsymbol{o}_i, \boldsymbol{x}_j)$. We minimize the following loss:

$$\mathcal{L}_{\text{info}} = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{\exp(f(\boldsymbol{o}_i, \boldsymbol{x}_i))}{\sum_{j=1}^{n} \exp(f(\boldsymbol{o}_i, \boldsymbol{x}_j))}. \tag{31}$$

Following the derivation by [Oord *et al.*, 2018], minimizing this loss is equivalent to maximizing a lower bound on MI:

$$\mathcal{I}(\boldsymbol{\mathcal{O}}; \boldsymbol{\mathcal{X}}) \geq \log(n) - \mathcal{L}_{\text{info}}. \tag{32}$$

To enforce spatial smoothness while respecting semantic heterogeneity (i.e., spatial coherence constraint), we further introduce an adaptive Laplacian regularization term $\mathcal{L}_{\text{sth}}$:

$$\mathcal{L}_{\text{sth}} = \frac{1}{|\boldsymbol{\mathcal{E}}|} \sum_{(i,j)\in\boldsymbol{\mathcal{E}}} \tilde{\boldsymbol{w}}_{ij} \cdot \|x_i - x_j\|_2^2, \tag{33}$$

where $\tilde{\boldsymbol{w}}_{ij} = \exp(-\|\boldsymbol{o}_i - \boldsymbol{o}_j\|_2)$ is the adaptive edge weight. Essentially, higher weights are assigned to neighbors with similar contexts, to enforce smoothness within functional regions and preserve boundaries between semantically distinct areas. The semantic alignment loss consists of two parts:

$$\mathcal{L}_{\text{sem}} = \mathcal{L}_{\text{info}} + \lambda \mathcal{L}_{\text{sth}}, \tag{34}$$

where λ is the trade-off between the two losses. Following previous efforts in traffic and flow forecasting [Song *et al.*, 2020; Xie *et al.*, 2023], we adopt the Huber loss as the training objective due to its robustness to outliers:

$$\mathcal{L}_{pred} = \begin{cases} \frac{1}{2}(\boldsymbol{Y} - \hat{\boldsymbol{Y}})^2 & \text{if } |\boldsymbol{Y} - \hat{\boldsymbol{Y}}| \leq \delta, \\ \delta|\boldsymbol{Y} - \hat{\boldsymbol{Y}}| - \frac{1}{2}\delta^2 & \text{otherwise}, \end{cases} \tag{35}$$

where δ is a threshold controlling the transition between quadratic and linear penalties, and the final loss is:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \alpha \mathcal{L}_{\text{sem}}, \tag{36}$$

where α balances the two loss terms, and additional algorithmic details are provided in Appendix A.

4 Experiments

Dataset. To validate the performance of SCM, we select two real-world metro flow datasets: Shanghai Metro [Sun *et al.*, 2025] and Hangzhou Metro [Liu *et al.*, 2020]. Following common preprocessing practices for metro flow, the raw flow data are aggregated into 10-minute intervals and normalized to zero mean [Xie *et al.*, 2023], resulting in six time steps per hour. For forecasting, metro flow observations from the past hour are used to predict the flow in the next hour, i.e., $\omega = 6$, consistent with previous studies [Xie *et al.*, 2023]. We adopt a standard data splitting strategy by dividing each dataset into training, validation, and testing sets with a ratio of 7:1:2. The statistics of the datasets are summarized in Table 1.

Dataset	#Node	#Edge	#Time step	#Interval
Shanghai Metro (SHMETRO)	80	84	8720	10 min
Hangzhou Metro (HZMETRO)	302	347	12546	10 min

Table 1: Statistics of the datasets.

Baselines. We evaluate SCM against 11 representative baselines, including widely used traffic flow forecasting models and recent metro-specific methods: DCRNN [Li *et al.*, 2017], STGCN [Yu *et al.*, 2017], AGCRN [Liu *et al.*, 2020], STSGCN [Song *et al.*, 2020], STTN [Xu *et al.*, 2020], Multi-SPANS [Zou *et al.*, 2024], TimeMixer [Wang *et al.*, 2024], ASTGCN [Guo *et al.*, 2019], PDFormer [Jiang *et al.*, 2023], STD-PLM [Huang *et al.*, 2025], ReDyNet [Gao *et al.*, 2025].
Implementations. SCM is implemented in PyTorch and accelerated by an NVIDIA RTX 4090 GPU. The order of Chebyshev polynomials is $k = 4$. The number of attention heads H is 4. The α is set to 0.3. The λ is 0.3. The batch size b is 16. We adopt the AdamW optimizer with an initial learning rate of 0.003. The model is trained for up to 300 epochs, with a step-wise learning rate decay strategy. More details are available at https://anonymous.4open.science/r/SCM-1212.
Metrics. We use three commonly used evaluation protocols: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).
Overall Performance. Table 2 shows the results of all methods, with the best result highlighted in bold and the second-best result underlined. We have the following findings:

Among baselines, models relying on predefined graph structures, such as DCRNN, STGCN and ASTGCN, achieve competitive performance in short-term forecasting but exhibit progressively larger errors as the forecasting horizon extends. The performance degradation indicates their limited capability in adapting to the dynamic and evolving interactions induced by time-varying metro flow patterns, as the fixed graph topology constrains the modeling of changing spatial dependencies. As the time horizon increases, models incorporating adaptive or dynamic graph mechanisms, including AGCRN and ReDyNet, consistently outperform static graph-based approaches across both datasets. These methods better capture the evolving spatiotemporal correlations, resulting in more stable performance in medium-term and long-term forecasting. PDFormer further improves forecasting accuracy by integrating attention-based spatial modeling, yet its reliance on structural priors still limits its flexibility. As a purely time-series modeling approach, TimeMixer does not explicitly incorporate graph structural information. In scenarios like metro systems, which rely on spatial topology, TimeMixer performs worse than GNN-based models. In contrast, our SCM achieves the best performance across all forecasting horizons, indicating that relying solely on predefined topologies or structural adaptation has limitations, while incorporating station semantic context better captures the dynamic spatiotemporal interactions within metro networks.
Ablation Study. To validate the effectiveness of the module design, we tested five variants in SCM: *w/o SL* (excludes the semantic alignment loss); *w/o MSC* (blocks the complete MSC module); *w/o SD_W* (replaces node-adaptive convolution in DSG with static and shared graph convolution); *w/o SD_S* (uses a fixed adjacency matrix instead of dynamic learn-

	Model	10 min			20 min			30 min			40 min			50 min			60 min		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Shanghai Metro	DCRNN	21.52	39.54	21.34%	22.29	41.17	22.04%	23.16	43.12	22.53%	24.24	45.15	23.75%	25.49	48.06	25.47%	27.00	51.51	32.07%
	STGCN	21.69	38.01	25.61%	23.10	41.34	29.74%	24.46	43.83	33.74%	26.04	46.85	39.72%	27.90	51.07	42.03%	29.93	56.13	51.28%
	AGCRN	18.52	34.39	17.12%	18.81	35.14	17.35%	19.08	35.78	17.94%	19.30	36.36	18.27%	19.46	36.72	18.47%	19.53	36.92	18.50%
	STSGCN	22.46	41.89	19.66%	22.54	42.23	19.73%	22.79	42.80	20.34%	23.10	43.49	20.78%	23.42	44.16	21.14%	23.81	44.94	21.68%
	STTN	23.21	44.83	21.99%	23.04	44.86	21.45%	23.34	45.80	21.86%	23.63	46.57	22.02%	24.42	47.13	21.89%	24.69	47.68	23.86%
	MultiSPANS	23.00	56.11	22.39%	22.70	56.28	22.80%	22.78	55.32	22.15%	22.91	52.89	21.91%	23.95	58.46	21.34%	23.59	49.92	22.38%
	TimeMixer	56.94	158.13	40.75%	57.12	159.38	41.54%	57.37	159.16	41.75%	57.84	159.94	42.29%	58.12	160.07	42.34%	58.55	160.64	44.60%
	ASTGCN	24.49	47.94	26.38%	25.07	47.80	27.49%	25.21	48.15	27.52%	26.23	58.28	27.94%	27.71	53.05	28.45%	38.83	57.84	30.89%
	PDFormer	18.61	35.12	16.77%	18.67	35.38	16.67%	18.81	35.85	16.78%	19.00	35.99	17.05%	19.38	36.78	17.35%	19.89	37.75	17.78%
	STD-PLM	20.18	36.85	18.57%	20.34	37.47	18.58%	20.53	37.98	18.66%	20.76	38.50	18.81%	21.04	39.10	18.99%	21.39	39.88	19.24%
	ReDyNet	18.62	34.24	16.38%	18.97	35.13	16.77%	19.09	35.65	17.10%	19.27	36.06	17.28%	19.47	36.54	17.45%	19.69	36.98	17.59%
	SCM	**18.33**	**33.78**	**16.20%**	**18.53**	**34.49**	**16.33%**	**18.68**	**34.91**	**16.43%**	**18.81**	**35.27**	**16.51%**	**18.94**	**35.59**	**16.60%**	**19.06**	**35.90**	**16.70%**
Hangzhou Metro	DCRNN	19.92	35.51	23.79%	28.78	37.73	26.34%	21.96	39.61	28.46%	24.84	44.46	30.96%	25.85	49.19	33.85%	27.84	53.89	37.59%
	STGCN	19.86	33.87	28.26%	20.62	34.75	31.26%	21.72	36.33	34.39%	23.72	40.62	38.45%	25.29	43.38	43.14%	27.38	46.79	49.85%
	AGCRN	18.92	32.45	24.23%	19.91	32.72	24.76%	19.27	33.23	24.99%	19.58	34.23	26.48%	20.22	35.81	28.81%	20.60	36.21	33.95%
	STSGCN	20.06	34.69	23.15%	20.23	35.01	22.86%	20.54	35.53	23.21%	21.82	36.46	24.06%	21.46	37.36	24.51%	21.94	38.41	25.01%
	STTN	21.02	36.48	24.21%	20.76	36.55	24.90%	21.02	36.92	26.28%	21.31	37.47	27.65%	21.37	37.30	30.64%	21.92	38.43	34.26%
	MultiSPANS	27.76	54.01	32.30%	26.76	52.31	33.56%	26.68	52.48	33.59%	27.32	53.40	35.89%	28.28	55.22	39.56%	30.53	59.73	43.37%
	TimeMixer	47.88	92.88	67.82%	47.58	91.67	78.71%	47.68	91.48	65.86%	48.40	93.19	67.96%	48.82	94.65	67.83%	49.69	95.86	69.28%
	ASTGCN	33.83	72.72	41.50%	33.10	78.63	49.13%	33.28	71.09	61.23%	35.42	74.77	82.01%	36.83	75.72	114.76%	37.87	78.39	186.59%
	PDFormer	18.92	32.45	24.23%	19.01	32.72	24.76%	19.27	35.23	24.99%	19.58	34.23	26.48%	20.22	35.88	28.01%	20.60	36.22	33.95%
	STD-PLM	20.97	37.23	22.91%	21.49	38.26	23.54%	22.03	39.07	24.10%	22.88	40.81	24.97%	23.77	42.71	25.95%	24.69	44.82	27.05%
	ReDyNet	17.89	30.54	20.42%	18.16	31.18	20.42%	18.37	31.62	20.46%	18.64	32.21	20.70%	18.85	32.63	21.04%	19.05	33.02	21.33%
	SCM	**17.39**	**28.97**	**19.18%**	**17.59**	**29.44**	**19.26%**	**17.75**	**29.78**	**19.39%**	**17.92**	**30.13**	**19.56%**	**18.07**	**30.45**	**19.67%**	**18.18**	**30.69**	**19.76%**

Table 2: Performance comparison of SCM and baselines on Shanghai Metro, Hangzhou Metro.

ing in DSG); *w/o SDG* (replaces the outputs of SDG with random embeddings); *w/o SD_SW* (simultaneously removes both the parameter pooling and dynamic adjacency matrix in DSG, degenerating into a fully static graph convolution). As shown in Fig. 3, we find that: (1) removing any module leads to a degradation in model performance. (2) *w/o SL* reduces the alignment between semantic representations and flow, leading to less stable performance. (3) *w/o MSC* limits the model's ability to capture flow patterns at different scales, yielding a performance drop as the horizon increases. (4) *w/o SDG* removes station semantic features, resulting in a notable performance drop, especially in long-term forecasting. (5) *w/o SD_SW* weakens the modeling of spatial dependencies, resulting in higher errors across all forecasting horizons.

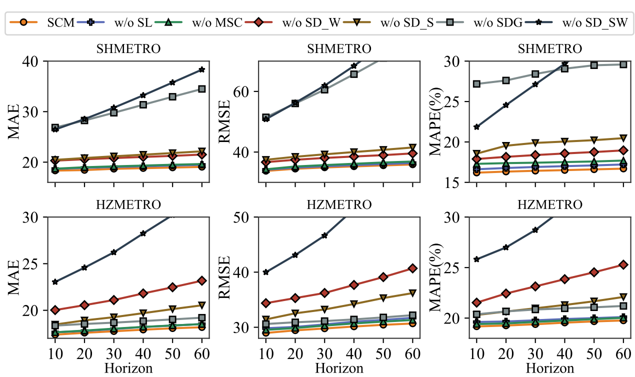

Figure 3: The ablation study on SHMETRO and HZMETRO.

External Context Impact. To evaluate the impact of embeddings obtained by our CSE, we consider three variants: SCM-ED (without date and weather embeddings), SCM-D (without date embedding), and SCM-E (without weather embedding). Table 3 shows that removing any embedding degrades performance, confirming their effectiveness. In particular, excluding the date embedding leads to the largest drop, underscoring its importance for capturing temporal regularities, while the weather embedding has a relatively minor impact.

Horizon	Metric	**SCM**	SCM-E	SCM-D	SCM-ED
10min	MAE	**17.39**	17.68	17.89	17.72
	RMSE	**28.97**	29.87	30.57	30.03
	MAPE	**19.18%**	19.76%	19.54%	19.50%
20min	MAE	**17.59**	17.92	18.12	17.97
	RMSE	**29.44**	30.41	31.16	30.90
	MAPE	**19.26%**	19.95%	19.65%	19.70%
30min	MAE	**17.75**	18.15	18.39	18.21
	RMSE	**29.78**	30.80	31.66	31.35
	MAPE	**19.39%**	20.20%	22.20%	19.93%
40min	MAE	**17.92**	18.41	18.69	18.49
	RMSE	**30.14**	31.48	32.25	31.94
	MAPE	**19.54%**	20.45%	20.03%	20.18%
50min	MAE	**18.07**	18.62	18.95	18.75
	RMSE	**30.45**	31.82	32.74	32.48
	MAPE	**19.67%**	20.66%	20.23%	20.42%
60min	MAE	**18.18**	18.80	19.20	18.97
	RMSE	**30.69**	32.13	33.19	32.96
	MAPE	**19.76%**	20.85%	20.50%	20.67%

Table 3: Ablation study on Hangzhou Metro.

Semantic Interpretability. To illustrate the interpretability of station semantics, we select a station located near a major leisure destination as an example. Fig. 4 shows the flow dynamics of a station that is semantically identified by the SDG as a leisure destination. This functional role is not predefined, but is automatically generated by SDG through the systematic organization of heterogeneous station-level information. Under this SDG-generated semantic context, the observed flow patterns become interpretable: weekend traffic exhibits a pronounced leisure-driven profile, characterized by an outflow peak around 10:00 and a delayed inbound peak in the late afternoon, corresponding to typical departure–return routines of recreational trips. In contrast, weekday flows are dominated by routine commuting and school-related travel, resulting in smoother temporal variations with less prominent peaks. By consolidating station and POI attributes into semantically meaningful representations and coupling them with temporal context in SCM, SDG enables functional-level explanations of flow variations, providing both stronger predictive grounding and enhanced semantic interpretability.

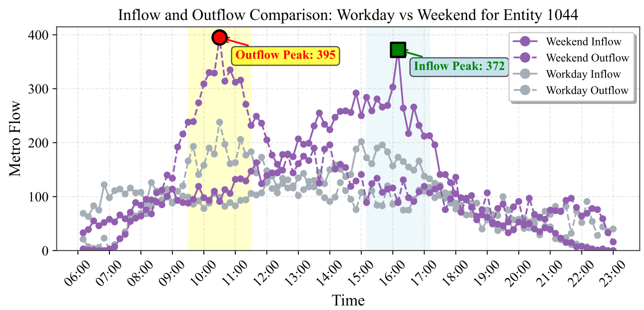

Figure 4: Flow for Weekend vs Workday at Leisure Destination.

Semantic-Consistent Interpretability. To further show the value of consistency with semantic descriptions, Fig. 5 shows the relationship between semantic spatial similarity and dynamic graph correlation for two stations of the Hangzhou Metro. The semantic representations generated by the SDG indicate that these two stations exhibit high functional consistency, both primarily serving residential and educational purposes, with surrounding environments dominated by residential areas and schools. Based on this semantic representation, DSG assigned high association strengths to these two stations during dynamic graph construction, reflecting their alignment in latent travel demand and metro flow patterns. In the actual metro system, stations with similar functional and spatial attributes often exhibit comparable fluctuation patterns during critical periods like rush hours. This result demonstrates that the semantic information generated by SDG effectively constrains DSG's structural learning, ensuring dynamic graph relevance aligns with station functional semantics.

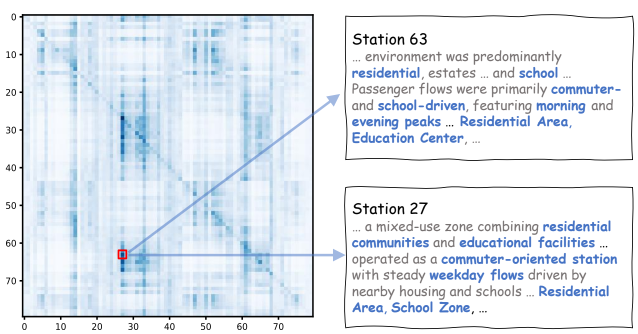

Figure 5: The semantic contrastive analysis on Hangzhou Metro.

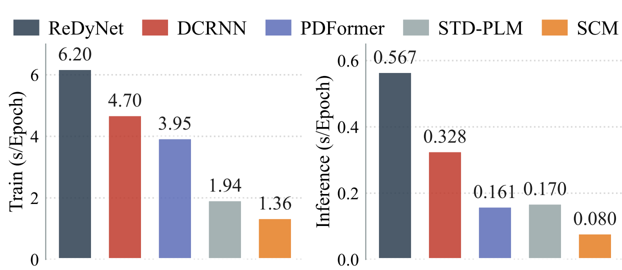

Figure 6: The efficiency analysis on Hangzhou Metro.

Efficiency. In the end, Fig. 6 presents the efficiency comparison on *Hangzhou Metro* showing that SCM achieves competitive training and inference costs. Additional experiments, including sensitivity analysis and extensions of SCM to traffic forecasting tasks, are provided in Appendix C.

5 Related Work

Metro Flow Forecasting. Recent progress in flow forecasting has focused on modeling complex spatiotemporal patterns, particularly for predicting traffic relationships in road networks [Wu *et al.*, 2020]. They employ various sequential modeling approaches such as LLM [Huang *et al.*, 2025] and attention mechanisms [Guo *et al.*, 2019; Jiang *et al.*, 2023]. Metro flow has become a vital component of urban transportation, attracting growing research interest in modeling passenger flow distributions across stations. However, compared with other traffic flows, metro flow exhibits unique characteristics, reflected in station-specific semantic patterns and their dynamically evolving inter-station relationships. These station-specific semantic embeddings make capturing semantic alignment between stations and their corresponding traffic data critically important. Recently, Gao et al. [Gao *et al.*, 2025] introduced ReDyNet, which captures the complex spatiotemporal dynamics of metro flow by dynamically adapting to variations in flow patterns and external influences, and disentangling redundant signals for improved forecasting accuracy. While ReDyNet has shown significant progress in capturing the spatiotemporal dynamics of metro flow, it still faces challenges in effectively modeling station-specific semantic information and accounting for complex contextual dependencies between stations.

Traffic Flow Application with LLM. Recent studies have explored the integration of LLM [Jin *et al.*, 2023] into traffic flow perspectives. One line of work treats LLMs as spatio-temporal encoders by tokenizing traffic signals and fine-tuning transformer backbones to model temporal dependencies, such as ST-LLM [Liu *et al.*, 2024], TPLLM [Ren *et al.*, 2024], and STD-PLM [Huang *et al.*, 2025], which employ partial fine-tuning or parameter-efficient strategies (e.g., LoRA) to adapt LLMs for numerical prediction. However, real-world traffic data often suffers from missing values and noise, motivating methods that combine LLMs with imputation mechanisms, including GATGPT [Chen *et al.*, 2023], which integrates graph attention with LLMs for missing data completion. Another direction investigates LLMs as predictors themselves. UrbanGPT [Li *et al.*, 2024] incorporates spatio-temporal encoders and context-enriched prompts for traffic prediction under limited data. While these studies demonstrate the potential of LLMs in traffic forecasting, most approaches either rely heavily on LLMs for prediction or use them as auxiliary semantic encoders, whereas effectively integrating stable station semantics with dynamic spatio-temporal interactions remains underexplored.

6 Conclusion

We introduced SCM, a novel semantic-aware dynamic graph framework for metro flow forecasting. In particular, our SCM generates structured station semantics via an LLM, models station-level interactions with dynamic dependency learning and multi-scale spatiotemporal convolutions, and enforces semantic–flow alignment through an information-theoretic semantic consistency objective with spatial coherence. Extensive experiments on two real-world metro datasets show that SCM consistently outperforms representative baselines.

References

[Bai *et al.*, 2020] Lei Bai, Lina Yao, Can Li, Xianzhi Wang, and Can Wang. Adaptive graph convolutional recurrent network for traffic forecasting. *Advances in neural information processing systems*, 33:17804–17815, 2020.

[Chen *et al.*, 2023] Yakun Chen, Xianzhi Wang, and Guandong Xu. Gatgpt: A pre-trained large language model with graph attention network for spatiotemporal imputation. *arXiv preprint arXiv:2311.14332*, 2023.

[Defferrard *et al.*, 2016] Michaël Defferrard, Xavier Bresson, and Pierre Vandergheynst. Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in neural information processing systems*, 29, 2016.

[Devlin *et al.*, 2018] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. *CoRR*, abs/1810.04805, 2018.

[Gao *et al.*, 2025] Qiang Gao, Zizheng Wang, Li Huang, Goce Trajcevski, Guisong Liu, and Xueqin Chen. Responsive dynamic graph disentanglement for metro flow forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 11690–11698, 2025.

[Guo *et al.*, 2019] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 922–929, 2019.

[Huang *et al.*, 2025] Yiheng Huang, Xiaowei Mao, Shengnan Guo, Yubin Chen, Junfeng Shen, Tiankuo Li, Youfang Lin, and Huaiyu Wan. Std-plm: Understanding both spatial and temporal properties of spatial-temporal data with plm. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 11817–11825, 2025.

[Jiang *et al.*, 2023] Jiawei Jiang, Chengkai Han, Wayne Xin Zhao, and Jingyuan Wang. Pdformer: Propagation delay-aware dynamic long-range transformer for traffic flow prediction. In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pages 4365–4373, 2023.

[Jin *et al.*, 2023] Ming Jin, Qingsong Wen, Yuxuan Liang, Chaoli Zhang, Siqiao Xue, Xue Wang, James Zhang, Yi Wang, Haifeng Chen, Xiaoli Li, et al. Large models for time series and spatio-temporal data: A survey and outlook. *arXiv preprint arXiv:2310.10196*, 2023.

[Lan *et al.*, 2022] Shiyong Lan, Yitong Ma, Weikang Huang, Wenwu Wang, Hongyu Yang, and Pyang Li. Dstagnn: Dynamic spatial-temporal aware graph neural network for traffic flow forecasting. In *International conference on machine learning*, pages 11906–11917. PMLR, 2022.

[Li *et al.*, 2017] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. *arXiv preprint arXiv:1707.01926*, 2017.

[Li *et al.*, 2024] Zhonghang Li, Lianghao Xia, Jiabin Tang, Yong Xu, Lei Shi, Long Xia, Dawei Yin, and Chao Huang. Urbangpt: Spatio-temporal large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 5351–5362, 2024.

[Liu *et al.*, 2019] Yang Liu, Zhiyuan Liu, and Ruo Jia. Deeppf: A deep learning based architecture for metro passenger flow prediction. *Transportation Research Part C: Emerging Technologies*, 101:18–34, 2019.

[Liu *et al.*, 2020] Lingbo Liu, Jingwen Chen, Hefeng Wu, Jiajie Zhen, Guanbin Li, and Liang Lin. Physical-virtual collaboration modeling for intra-and inter-station metro ridership prediction. *IEEE Transactions on Intelligent Transportation Systems*, 2020.

[Liu *et al.*, 2024] Chenxi Liu, Sun Yang, Qianxiong Xu, Zhishuai Li, Cheng Long, Ziyue Li, and Rui Zhao. Spatial-temporal large language model for traffic prediction. In *2024 25th IEEE International Conference on Mobile Data Management (MDM)*, pages 31–40. IEEE, 2024.

[Oord *et al.*, 2018] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

[Ren *et al.*, 2024] Yilong Ren, Yue Chen, Shuai Liu, Boyue Wang, Haiyang Yu, and Zhiyong Cui. Tpllm: A traffic prediction framework based on pretrained large language models. *arXiv preprint arXiv:2403.02221*, 2024.

[Song *et al.*, 2020] Chao Song, Youfang Lin, Shengnan Guo, and Huaiyu Wan. Spatial-temporal synchronous graph convolutional networks: A new framework for spatial-temporal network data forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 914–921, 2020.

[Sun *et al.*, 2025] Peiyan Sun, Jinming Yang, Zongyuan Huang, Shaoyu Huang, Shengyuan Xu, Weipeng Wang, Wenxuan Guo, Yuting Feng, Xi Zhai, Tao Yang, et al. Human mobility datasets in the complex metro system of shanghai. *Scientific Data*, 12(1):1061, 2025.

[Wang *et al.*, 2024] Shiyu Wang, Haixu Wu, Xiaoming Shi, Tengge Hu, Huakun Luo, Lintao Ma, James Y Zhang, and Jun Zhou. Timemixer: Decomposable multiscale mixing for time series forecasting. *arXiv preprint arXiv:2405.14616*, 2024.

[Wu *et al.*, 2020] Zonghan Wu, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, and S Yu Philip. A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems*, 32(1):4–24, 2020.

[Xie *et al.*, 2023] Peng Xie, Minbo Ma, Tianrui Li, Shenggong Ji, Shengdong Du, Zeng Yu, and Junbo Zhang. Spatio-temporal dynamic graph relation learning for urban metro flow prediction. *IEEE Transactions on Knowledge and Data Engineering*, 35(10):9973–9984, 2023.

[Xiong *et al.*, 2019] Zhi Xiong, Jianchun Zheng, Dunjiang Song, Shaobo Zhong, and Quanyi Huang. Passenger flow prediction of urban rail transit based on deep learning methods. *Smart Cities*, 2(3):371–387, 2019.

[Xu *et al.*, 2020] Mingxing Xu, Wenrui Dai, Chunmiao Liu, Xing Gao, Weiyao Lin, Guo-Jun Qi, and Hongkai Xiong. Spatial-temporal transformer networks for traffic flow forecasting. *arXiv preprint arXiv:2001.02908*, 2020.

[Yao *et al.*, 2019] Liang Yao, Chengsheng Mao, and Yuan Luo. Graph convolutional networks for text classification. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 7370–7377, 2019.

[Yu *et al.*, 2017] Bing Yu, Haoteng Yin, and Zhanxing Zhu. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. *arXiv preprint arXiv:1709.04875*, 2017.

[Zhu *et al.*, 2024] Li Zhu, Cheng Chen, Hongwei Wang, F. Richard Yu, and Tao Tang. Machine learning in urban rail transit systems: A survey. *IEEE Transactions on Intelligent Transportation Systems*, 25(3):2182–2207, 2024.

[Zou *et al.*, 2024] Dongcheng Zou, Senzhang Wang, Xuefeng Li, Hao Peng, Yuandong Wang, Chunyang Liu, Kehua Sheng, and Bo Zhang. Multispans: A multi-range spatial-temporal transformer network for traffic forecast via structural entropy optimization. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, WSDM '24, page 1032–1041, New York, NY, USA, 2024. Association for Computing Machinery.